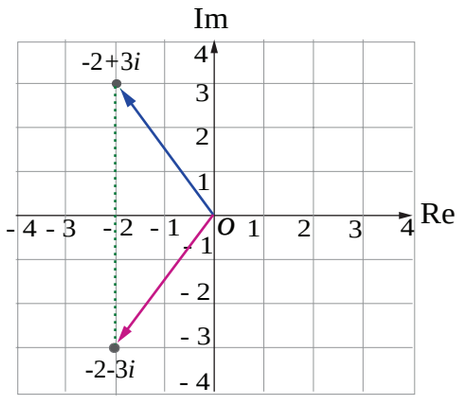
The complex conjugate of z is denoted by $\overline{z}$. To get the conjugate of the complex number z, simply change i by $-i$ in z. For instance $2-5i$ is the conjugate of $2+5i$. The product of a complex number with its conjugate is a real number.

For instance, (i) $(x+iy)(x-iy) = x^2 - (iy)^2 = x^2 + y^2$

(ii) $(1+3i)(1-3i) = (1)^2 - (3i)^2 = 1+9 = 10$.

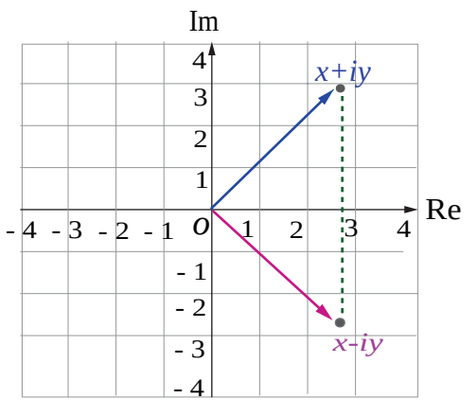
Geometrically, the conjugate of z is obtained by reflecting z on the real axis.

2.4.1 Geometrical representation of conjugate of a complex number



conjugate of a complex number

Fig. 2.14



conjugate of a complex number

Fig. 2.15

Note

Two complex numbers $x+iy$ and $x-iy$ are conjugates to each other. The conjugate is useful in division of complex numbers. The complex number can be replaced with a real number in the denominator by multiplying the numerator and denominator by the conjugate of the denominator. This process is similar to rationalising the denominator to remove surds.

2.4.2 Properties of Complex Conjugates

(1) $\overline{z_1 + z_2} = \overline{z_1} + \overline{z_2}$

(2) $\overline{z_1 - z_2} = \overline{z_1} - \overline{z_2}$

(3) $\overline{z_1 z_2} = \overline{z_1}\,\overline{z_2}$

(4) $\overline{\left(\frac{z_1}{z_2}\right)} = \frac{\overline{z}_1}{\overline{z}_2}$, $z_2 \neq 0$

(5) $\mathrm{Re}(z) = \frac{z+\overline{z}}{2}$

(6) $\mathrm{Im}(z) = \frac{z-\overline{z}}{2i}$

(7) $\overline{(z^n)} = (\overline{z})^n$, where n is an integer

(8) z is real if and only if $z = \overline{z}$

(9) z is purely imaginary if and only if $z = -\overline{z}$

(10) $\overline{\overline{z}} = z$

Let us verify some of the properties.

Property

For any two complex numbers z_1 and z_2, we have $\overline{z_1 + z_2} = \overline{z_1} + \overline{z_2}$.

Proof

Let $z_1 = x_1 + iy_1$, $z_2 = x_2 + iy_2$, and x_1, x_2, y_1, and $y_2 \in \mathbb{R}$

$$\overline{z_1 + z_2} = \overline{(x_1 + iy_1) + (x_2 + iy_2)}$$